

BiasClean External Validation in the Justice Domain

Phase 2 Real-Dataset Validation Report

AI Fairness CIO

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Pipeline version validated against: 3.6.9 (biasclean_v3_5_1_terminal.py)

1. Executive Summary

This report documents Phase 2 of the BiasClean™ Justice-domain external validation: running the pipeline against six real, publicly sourced justice datasets under a single consistent policy; every run accepted the pipeline's own auto-detected defaults, simulating exactly how a non-technical user operating the no-code tool would use it. No target column, protected-feature mapping, or column exclusion was ever hand-picked to produce a better-looking result. Where auto-detection produced a wrong or misleading outcome, the fix was made in the pipeline's code and verified before the affected dataset was re-run, per the SOP's root-cause-first protocol.

Dataset	Status	Headline result
COMPAS (ProPublica)	Validated	0.095 → 0.051; SVM rejected for new reversals
NC Statewide Traffic Stops	Validated	0.067 → 0.031; regulator flagged unresolved reversals in 3 of 4 features
NIJ Recidivism Challenge 2021	Validated	0.047 (unchanged, below mitigation threshold)
Oklahoma City Traffic Stops	Validated	0.0003, technically correct, minimal real signal
UCI Communities and Crime	Validated	0.073 → 0.027; 89% of Region groups too small to rebalance
opafy25nid.csv	Excluded	No data dictionary located; source unidentified

Three previously-latent pipeline bugs were found and fixed in the course of this validation (v3.6.7, v3.6.8, v3.6.9), each traced to a specific real dataset, root-caused to an exact code path, fixed, covered by a regression test, and verified against the real data that surfaced it before being treated as resolved. Full detail in Section 5.

The single most consequential finding across all six datasets is North Carolina's rebalancing result: the pipeline's own fairness regulator flagged unresolved overcorrection in Ethnicity, Region, and Age, including a reversal implicating Wake County, North Carolina's second-most-populous county, and explicitly declined to certify these rates as safe to use without manual review. Full detail in Section 3.2.

2. Methodology

All six datasets were validated against the same version-evolving copy of the pipeline (biasclean_v3_5_1_terminal.py), progressing from v3.5.1 through v3.6.9 as bugs were found and fixed over the course of this validation round. Every dataset's final reported result in this document reflects v3.6.9 unless explicitly noted otherwise.

- Non-technical-user policy: at every interactive prompt (target column, auto-approval threshold, domain, mode), the default was accepted, pressing Enter, not typing an override. This was a deliberate choice: BiasClean is marketed as a no-code tool, so its validation needed to test what an unassisted user actually experiences, not an idealized expert-configured run.
- Mode: Legacy (v2.7-equivalent rebalancing behavior) was used throughout for consistency across datasets. SVM fairness enforcement was enabled where computationally feasible; disabled for North Carolina after an initial attempt approached memory exhaustion at that dataset's scale (20.3 million rows) without completing.
- Root-cause-first protocol: per the SOP, every anomaly was traced to its exact cause before being classified as either a dataset limitation (accepted and documented) or a pipeline defect (fixed in code, tested, and the affected dataset re-validated).
- Freshness discipline: every result in this report was cross-checked against its own audit_trail.json and report.pdf pair, generated within this validation round; no result is carried over from an earlier pipeline version without being explicitly re-confirmed.

2.1 The Seven-Feature Weighting Framework

BiasClean assigns structured, evidence-based weights to seven fairness features per domain, using a three-stage framework: Structural Inequality Weight (SIW), Evidence Strength Weight (ESW), and Policy & Legal Relevance Weight (PLW). Features were scored on SIW, ESW, and PLW and normalised so domain totals equal 1.00, per *BiasClean v2.0, Evidence Base & Methodology Appendix (UK Edition)* (Tavakoli, internal, 2026).

This weighting is UK-specific. The underlying evidence base (ONS Census and labour market statistics, Ministry of Justice race and CJS reports, EHRC guidance, and others) and legal grounding (Equality Act 2010) reflect UK structural inequality patterns and regulatory priorities. Five of this report's six datasets are US-sourced; applying UK-derived weights to them means the *relative prioritization* across features, for example, Ethnicity weighted at 0.25 versus Gender at 0.05 for Justice, reflects UK evidence, not necessarily the evidence base or legal framework of the jurisdiction each dataset was drawn from. The pipeline's mechanical safeguards (rebalancing, the fairness regulator, small-group floors, reversal detection) operate independently of the specific weight values and are not affected by this, but the *composite bias score* and any weight-driven ranking of which feature matters most should be read with this in mind for non-UK data.

Feature	Justification
Region	Postcode-linked opportunity and service inequality.
Ethnicity	Strongest structural disparity across multiple domains.
Age	Behavioural gradients affecting outcomes.
Gender	Documented bias across hiring, health and leadership.
DisabilityStatus	Protected characteristic with consistent disadvantage.
SocioeconomicStatus (SES)	Key driver of multi-domain deprivation.
MigrationStatus	Affects service access, representation and civic participation.

For the Justice domain specifically, weights are: Ethnicity 0.25 (highest), SocioeconomicStatus 0.20, Region 0.15, Age 0.15, MigrationStatus 0.10, DisabilityStatus 0.10, Gender 0.05. This ordering is Justice-specific, not universal; Finance and Education weight SocioeconomicStatus highest (0.30 and 0.25 respectively), not Ethnicity.

Feature	Weight
Ethnicity	0.25
SocioeconomicStatus	0.20
Region	0.15
Age	0.15
MigrationStatus	0.10
DisabilityStatus	0.10
Gender	0.05

3. Per-Dataset Results

3.1 COMPAS, ProPublica Recidivism Dataset

Target column	is_recid
Records	7,214 → 7,238 (100.3% retention)
Bias score	0.0948 → 0.0505 (rebalanced stage; +47%)
SVM stage	0.0129, REJECTED by gate

Rebalance gate	Accepted (no actionable reversal)
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Protected features

Feature	Column used	Weight	p-value	Disparity: before → after
Gender	sex	0.05	3.28e-17	14.2% → 12.6%
Age	age_cat	0.15	8.67e-46	22.0% → 2.3%
Ethnicity	race	0.25	1.38e-31	21.9% → 16.3%

Finding: SVM enforcement was correctly rejected by the pipeline's own gate for introducing new disadvantaged-group reversals in Gender and Ethnicity that were not present after simple rebalancing; the final result correctly falls back to the rebalanced stage rather than the (lower-looking, but less trustworthy) SVM score.

Finding: Age's apparent group change (Greater than 45 → Less than 25) after rebalancing is correctly classified as near-parity, not an actionable reversal; the gap narrowed to 2.5 percentage points, below the feature's own 7.2-point disparity threshold.

Caveat: Two small Ethnicity groups (Asian, n=32; Native American, n=18) fall below the 50-sample rebalancing floor and were excluded from automatic correction, flagged for manual review rather than silently rebalanced.

Caveat: Mapping ties exist for both Age (age vs. age_cat, both auto-approved at 95%/90%) and the outcome column (is_recid vs. is_violent_recid vs. two_year_recid, all tied at 90%). The winning columns (age_cat, is_recid) are the intended, correct choices, but this was independently re-confirmed on this v3.6.9 re-run, not assumed to carry over from the original baseline.

This result was independently re-confirmed on v3.6.9 and is numerically identical to the original v3.5.1 baseline to the full decimal, closing out any question of whether the pipeline fixes made elsewhere in this validation round affected COMPAS's result. They did not.

3.2 North Carolina Statewide Traffic Stops

Target column	citation_issued (auto-detected)
Records	20,286,645 → 20,287,857 (100.0% retention)
Bias score	0.0670 → 0.0314 (+53.2%)
SVM stage	Not run (disabled, see note below)
Rebalance gate	REJECTED, flagged for manual review

Protected features

Feature	Column used	Weight	p-value	Improvement
Region	county_name	0.15	<0.001	+72.5%
Age	subject_age	0.15	<0.001	+50.2%
Ethnicity	subject_race	0.25	<0.001	+34.4%
Gender	subject_sex	0.05	<0.001	-136.6% (trivial magnitude)

Caveat: SVM enforcement was disabled for this run. An earlier attempt with SVM enabled approached ~2 hours of runtime and 33GB memory usage (system free memory fell to ~2.3GB) without completing, at this dataset's scale (20.3 million rows, 5+ million SVM validation samples). It was terminated rather than risk an OS-level kill. The rebalancing-stage result below completed successfully in both attempts and is unaffected by this decision.

Regulator-flagged reversals, the headline finding of this validation round

The fairness regulator explicitly declined to certify the rebalanced rates as safe in three of North Carolina's four protected features. These three reversals are not equivalent in severity:

Finding: Region: the disadvantaged county flipped from Granville County to Wake County (gap 14.4 percentage points after rebalancing). Wake County is North Carolina's second-most-populous county (~1.96 million stops in this dataset); this is a substantive result, not a small-sample artifact, and is the most consequential single finding in this validation round.

Finding: Ethnicity: the disadvantaged group flipped from 'black' to 'other' (gap 8.8 percentage points after rebalancing). 'Other' is a real 160,899-person group, not a sample-size artifact.

Caveat: Age: the disadvantaged group flipped from age 15 to age 103 (gap 49.6 percentage points after rebalancing). This is best understood as a methodological artifact, not a real age-103 effect: age 103 has only 10 samples in the full 20.3-million-row dataset and was explicitly excluded from automatic rebalancing for falling below the 50-sample floor. Its rate was frozen while every other age group's rate shifted toward the population mean during rebalancing, making the frozen group appear newly "disadvantaged" by comparison, without anything actually happening to it. This should not be read as equivalent in kind to the Region or Ethnicity reversals above.

Ages 100–109 (n = 1 to 42 each) were excluded from automatic rebalancing on the same basis.

Mapping resolution

Region resolved correctly to county_name over location (13,662 near-street-level groups, carrying a small-group warning), confirming the v3.6.7 tie-break fix on real 20-million-row data. Ethnicity resolved correctly to subject_race over the redundant raw_Race/raw_Ethnicity source columns. The pipeline's own SVM-stage statistical leakage detector (pre-existing, independent of this validation's fixes) automatically excluded 'outcome' and 'raw_action_description' as near-100%-purity target proxies during the earlier SVM attempt.

Finding: citation_issued (66% positive rate) produced a substantively rich, well-populated result across all four features, resolving an open question from earlier in this validation as to whether the auto-detected target would be informative enough to report on; it was. A separate run against search_conducted (originally considered as a richer, rarer-event alternative) was not required.

3.3 NIJ Recidivism Forecasting Challenge 2021

Target column	Recidivism_Within_3years (auto-detected)
Records	25,835 → 25,835 (100% retention; no rebalancing triggered)
Bias score	0.0468 (unchanged)
SVM stage	Not applied, mitigation skipped

Protected features

Feature	Column used	Weight	p-value	Group rates
Age	Age_at_Release	0.15	2.85e-175	18–22: 72.0% vs. 48+: 41.9% (30.1pp)
Gender	Gender	0.05	9.14e-49	M: 59.4% vs. F: 45.5% (13.9pp)
DisabilityStatus	Condition_MH_SA	0.10	1.82e-74	Yes: 61.7% vs. No: 49.9% (11.9pp)
Ethnicity	Race	0.25	1.72e-4	BLACK: 58.7% vs. WHITE: 56.3% (2.3pp)

Finding: The composite bias score (0.0468) stayed unchanged after rebalancing correctly; it sits under the 0.05 mitigation threshold even though all four individual features are statistically significant. This is the pipeline's intended dual-gate behavior (per-feature significance and composite-score threshold are evaluated separately), not a defect.

Finding: Target-column auto-detection correctly self-corrected to Recidivism_Within_3years, ranking it above four Violations_* columns that a prior pipeline version had mistakenly preferred, confirming a fix made earlier in this codebase's history (v3.6.2→v3.6.3) still holds.

Caveat: Two residual, unresolved limitations were concretely confirmed in this single run, both already disclosed in the pipeline's own change history as open questions rather than solved problems: (1) Recidivism_Within_3years beat three other same-pattern Recidivism_Arrest_YearN columns (all tied at identical confidence) purely by appearing first in the file, correct here, but by file-layout luck, not a guarantee. (2) DisabilityStatus resolved to Condition_MH_SA over Condition_Cog_Ed and Condition_Other, three columns tied at identical confidence representing genuinely different disability sub-types (mental health/substance abuse vs. cognitive/education vs. other) rather than redundant duplicates. The domain's weight framework allocates only one slot for DisabilityStatus, so this is a real modeling limitation, not just a mechanical tie to resolve.

3.4 Oklahoma City Traffic Stops

Target column	outcome (arrest=0 / citation=1, auto-detected)
Records	945,107 → 945,107 (100% retention; no rebalancing triggered)
Bias score	0.0003 (unchanged)
SVM stage	Not applied, mitigation skipped

Protected features

Feature	Column used	Weight	p-value	Note
Ethnicity	subject_race	0.25	1.49e-29	Trivial 0.1–0.3pp gaps
Gender	subject_sex	0.05	2.60e-27	Trivial 0.1–0.3pp gaps
Age	subject_age	0.15	1.00	Zero variance across all 98 groups
Region	vehicle_registration_state	0.15	1.00	Zero variance across all 51 states

Caveat: outcome's target distribution is 99.8% positive; essentially every stop in this jurisdiction's data ends in a citation regardless of who was stopped. Age and Region show literally zero variance ($p = 1.0$, the strongest possible "no relationship" result); Ethnicity and Gender register as statistically significant only because $n = 945,107$ gives enormous power to detect even a trivial 0.1–0.3 percentage-point difference. The composite score is technically accurate but reflects very little real signal.

Finding: This is a genuine data-richness limitation, not a target-selection mistake: Oklahoma City's file (confirmed against its full 27-column list) contains no search/frisk/discretion-point column analogous to North Carolina's search_conducted. The only candidates, outcome, arrest_made, citation_issued, are all near-duplicates of the same extreme split. outcome is the genuinely best available target for this file; the near-null result is an accurate characteristic of this jurisdiction's data, not a pipeline defect.

Finding: Region correctly resolved to vehicle_registration_state (51 groups) over location (204,447 near-unique values across 945,107 rows), confirming the v3.6.9 cardinality guard on the real dataset that originally surfaced this bug (see Section 5.3). Ethnicity correctly resolved to subject_race over the redundant raw_dfnd_race source column.

3.5 UCI Communities and Crime

Target column	HighCrime (pre-engineered binarization)
Records	1,994 → 1,994 (100% retention)
Bias score	0.0734 → 0.0268 (+63%)
SVM stage	0.0268 , ACCEPTED by gate

Rebalance gate	Accepted (no actionable reversal)
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Protected features

Feature	Column used	Weight	p-value	Disparity: before → after
Region	state	0.15	1.2e-72	49.0% → 38.9%

Caveat: Only Region was mapped in this dataset; Gender, Age, and Ethnicity were not, and this is a genuine domain-fit limitation rather than an auto-detection failure. This dataset's demographic columns (e.g., racepctblack, agePct12t21) are continuous per-community percentages, not individual-level categories, so the pipeline's categorical group-rate comparison framework cannot use them without a manual binarization step that was intentionally not performed, per the non-technical-user policy.

Finding: 31 of 35 state groups (~89%) fall below the 50-sample rebalancing floor and were excluded from automatic correction; the most severe small-group case across this entire validation, far exceeding COMPAS's comparable caveat (2 of 6 groups excluded). State '23' (17 samples, the most disadvantaged group both before and after) remained frozen at a 94.1-percentage-point gap through rebalancing specifically because it is floor-excluded; the SVM stage, which is not subject to the same floor, still meaningfully narrowed this to 42.0 percentage points.

Finding: Region correctly resolved to state over county (the tie that originally surfaced the v3.6.7 bug, see Section 5.1); county has 108 groups with min_group_size = 1 and p = 0.126 (not significant, matching UCI's own documentation flagging county as sparse and non-predictive); state has 46 groups (matching real US states) and p = 1.2e-72 (extremely significant).

4. Excluded Dataset: opafy25nid.csv

This file (66,662 rows × 22,149 columns) could not be validated and is excluded from the findings above. Its column names, POOFFICE, ZONE, SORGDL, TTSC1_1 through TTSC1_111, TTSC2_1 through TTSC2_59+, DAFROM1-17, RETEXT1-26, MWEIGHT, PCNTDEPT, MNTHDEPT, ENCRYPT1-2, are consistent with a wide-format administrative export where per-episode data has been flattened into one row per person with over a hundred repeated columns, most populated for only a handful of people. No data dictionary or source documentation for this file was located, including a targeted web search for the filename and column names.

Auto-detection reflected this accurately rather than papering over it: across all 22,149 columns, only one, a column literally named AGE, matched any recognizable protected-feature pattern. Two successive auto-detected target columns (TTSC1_60, then ACCAP after a pipeline fix) were both technically valid two-category columns but semantically unidentified; there is no way to know what either represents or whether it constitutes a meaningful bias-relevant outcome.

Caveat: This dataset remains parked pending my locating source documentation or a data dictionary. No result from it should be treated as a validated finding.

5. Pipeline Bugs Found During This Validation Round

Each bug below was root-caused to an exact code path, fixed, covered by a new regression test, and verified against the real dataset that surfaced it, per the SOP's bug-handling protocol. Two additional fixes from earlier in this pipeline's history (v3.6.2→v3.6.3 outcome-pattern specificity; v3.6.5→v3.6.6 self-comparison report text) were re-confirmed as still holding during this validation but predate this round and are not detailed here.

5.1 v3.6.7, Protected-feature mapping tie-break ignored confidence and validation quality

Found on: UCI Communities and Crime (county incorrectly beat state for Region); previously disclosed as a known-but-unfixed issue on Oklahoma City (raw_dfnd_race beat subject_race for Ethnicity).

Root cause: when multiple raw columns mapped to the same protected feature, the step that picks one column per feature was a plain dict-overwrite loop; whichever column was processed last silently won, with mapping confidence and each candidate's own validation statistics (already computed, but never consulted) both ignored entirely.

Fix: added a shared selection method ranking candidates by (no small-group-size warning first, then higher confidence, then first-encountered for genuine ties), used consistently everywhere a feature-to-column decision is made.

Verified: unit tests reproducing both real-data shapes, plus a full end-to-end synthetic pipeline run confirming the correct column wins.

5.2 v3.6.8, Target columns with almost no data were accepted as valid

Found on: opafy25nid.csv, the auto-detected target TTSC1_60 had only 2 non-null values across 66,662 rows, yet was accepted as a fully valid binary target, producing a NaN-scored report that still looked complete (charts, a Deployment Decision section, everything).

Root cause: the function responsible for reducing a target column to clean 0/1 checked that a column had exactly two distinct non-null values, but never checked how many non-null values existed in total. This gap existed across all three of that function's return paths (boolean columns, already-clean 0/1 numeric columns, and the general two-value mapping case); a first fix attempt covered only the last path, and a regression test written specifically to probe the other two caught the gap before it shipped.

Fix: added a coverage guard requiring the smaller of the two classes to have at least 50 samples, the same floor already used elsewhere in the pipeline for protected-attribute group sizes, applied uniformly across all three code paths.

Verified: synthetic reproduction of the exact scenario now correctly raises an actionable error instead of producing a result; confirmed no previously validated dataset's real target column is affected.

5.3 v3.6.9, Near-unique-per-row columns could be auto-approved as protected features

Found on: Oklahoma City, the location column (204,447 distinct values across 945,107 rows, essentially unique per row) tied with vehicle_registration_state for Region and won, causing the pipeline to grind for roughly two hours computing per-group statistics across hundreds of thousands of near-singleton groups before the run was terminated rather than risk exhausting system memory.

Root cause: the v3.6.7 fix ranks tied candidates by whether they carry a small-group-size warning, but both location and vehicle_registration_state carried that warning, one catastrophically (204,447 near-singleton groups), the other for a single rare out-of-state plate, and a binary warning flag cannot distinguish severity. With both candidates tied on warning status and confidence, selection fell back to file order, where location happened to appear first.

Fix: the auto-approval step now rejects any candidate whose group count exceeds total records divided by 50, a bound derived directly from the pipeline's existing 50-sample floor, not an arbitrary cutoff: more groups than that mathematically guarantees most groups are undersized. This rejects the problem column before the expensive per-group computation ever runs, addressing both the wrong-column selection and the runtime cost at the same root.

Verified: unit tests reproducing the exact real-data tie; a scaled end-to-end pipeline run (same ratio of unique-per-row to healthy-category columns) completing in 3 seconds with the correct column selected; confirmed no previously validated dataset's real Region column is affected.

6. Cross-Dataset Patterns

Small-group exclusion severity varies enormously by dataset

Dataset	Feature	Groups below 50-sample floor
COMPAS	Ethnicity	2 of 6 groups (Asian, Native American)
Communities and Crime	Region	31 of 35 groups (~89%)
North Carolina	Age	10 of ~90 age values (ages 100–109)

Communities and Crime's exclusion rate is far more severe than any other dataset in this validation, a direct consequence of spreading 1,994 records across 46 state-level groups, versus COMPAS's 6 ethnicity groups or North Carolina's tens of millions of rows. Small-group exclusion should be read relative to each dataset's own scale, not as a uniform severity signal.

The same-pattern-tie residual limitation remains open

Every fix in this validation round improved tie-breaking between columns matching different patterns or carrying different validation quality. None of them resolve a genuine tie between multiple columns matching the identical pattern at identical confidence with identical validation quality; NIJ's four Recidivism_Arrest_YearN/Within_3years columns and three DisabilityStatus Condition_* columns are both real, currently-open instances of this. Selection in that specific case still falls back to file order as a documented last resort, not a guarantee, and any future dataset that hits it should be flagged the same way rather than silently trusted.

Regulator-flagged reversals by dataset

Dataset	Reversals flagged	Severity
North Carolina	Ethnicity, Region, Age (3 of 4)	High, Region involves NC's 2nd-largest county
COMPAS	Gender, Ethnicity (SVM stage only)	Moderate, correctly caught and rejected by gate
Communities and Crime	None	None
NIJ	None	None
Oklahoma City	N/A, mitigation never triggered	N/A

North Carolina stands alone as the dataset where the regulator flagged multiple rebalancing-stage reversals as unresolved; every other dataset's reversals, where present, were either correctly caught and rejected at the SVM stage (COMPAS) or did not occur at all.

Target-column richness is a property of the dataset, not the pipeline

Oklahoma City's near-null result and North Carolina's substantively rich one both used auto-detected targets and correct pipeline behavior; the difference is that North Carolina's file contains a genuine officer-discretion column structure (search, frisk, arrest, citation, warning as distinct fields) while Oklahoma City's does not. A future dataset with a similarly narrow schema should be expected to produce a similarly thin result, independent of any further pipeline fixes.

7. Recommendation

Five of six datasets are validated against v3.6.9 with root-caused explanations for every anomaly encountered; the sixth (opafy25nid.csv) is honestly excluded rather than forced through. Three real, previously latent defects were found and fixed during this round, each with a regression test and real-data verification, not just a synthetic reproduction.

- Phase 3 (Fairlearn benchmark comparison) readiness: the fixed pipeline is ready to proceed, contingent on the residual limitations in Section 6 being carried forward as known, documented constraints rather than treated as fully closed.
- North Carolina's rebalancing output (Ethnicity, Region, Age) should not be used downstream, for training, reporting, or otherwise, without the manual review the pipeline's own regulator explicitly requested.

- SVM enforcement was exercised in only two of five validated datasets in this round (COMPAS: rejected; Communities and Crime: accepted); three datasets skipped it entirely, either by necessity at scale (North Carolina) or because rebalancing was never triggered (NIJ, Oklahoma City). Real-data coverage of the SVM path is thinner than the rebalancing path from this round alone and would benefit from targeted follow-up.
- opafy25nid.csv should remain excluded from any production or governance claims until its source is documented; the pipeline's current behavior on it (an honest failure or a low-confidence single-feature result, depending on version) is now correct and should not be mistaken for validation.

Full machine-readable results (audit_trail.json) and formatted reports (report.pdf) for every dataset referenced in this document are retained alongside the corrected datasets they were generated from, per the SOP's own record-keeping requirements.